Sixth powers in a mean square: an accessible guide to the 11/12 argument for $\zeta(s)$

Claude Opus 5.5 and OpenAI Astra

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A note from Michael Rubinstein. These are working notes, generated by Claude and Codex to help me read the papers on the quasi-Riemann hypothesis that OpenAI released.

Abstract

We give an accessible account of an argument showing that $\zeta(s)$ has no zeros in $\Re s > 11/12$. The averages are written out in classical terms: Möbius sums over Eisenstein integers, twisted by sextic residue symbols and averaged in mean square. Rows that are sixth powers recover the coefficients of $1/(\zeta(s)L(s,\chi_{-3}))$, and the mean square is bounded using Poisson summation, Kubota's cubic theta function, a quadratic large sieve, and a descent.

The proof in six steps.

1. To show that $\zeta(s)$ has no zeros with real part above $11/12$, it is enough to prove that the smoothed Möbius sum $A_1(D) = \sum_n \mu_K(n) W(\mathrm{N}(n)/D)$ over Eisenstein integers, whose coefficients are those of $1/(\zeta(s)L(s,\chi_{-3}))$, is at most about $D^{11/12}$.

2. That sum is embedded in a large family of similar sums $A_u(D) = \sum_n \mu_K(n)(u/n)_6 W(\mathrm{N}(n)/D)$, each twisted by a sextic residue symbol, and the roughly $H^{1/6}$ rows where $u = \pi^6$ is the sixth power of a prime reproduce the original sum, up to a small error.

3. The mean square $\sum_{0<\mathrm{N}(u)\le H} |A_u(D)|^2$ is a sum of nonnegative terms, so a bound of size about DH for it (square-root cancellation in each row, on average) controls the roughly $H^{1/6}$ copies of $|A_1(D)|^2$ it contains; this gives $|A_1(D)|^2 \lesssim DH^{5/6}$, and taking H a little larger than D gives the exponent $11/12$.

4. Poisson summation in the row variable u turns this average into a dual family with fewer rows, and here is the key trick: the Gauss sums produced by Poisson summation combine with the Möbius function to give *cubic* Gauss sums, because the cube of a cubic Gauss sum at a prime carries exactly a factor -1, which is the Möbius sign.

5. Cubic Gauss sums are, up to fixed residue-class factors, the Fourier coefficients of Kubota and Patterson's cubic theta function, whose symmetry under a group of matrices shortens each twisted sum and turns the sextic twist into an ordinary quadratic one, so that Heath-Brown's quadratic large sieve gives the needed bound.

6. The theta function's coefficients include extra terms indexed by cubes; these are removed by Möbius inversion and by a repeated two-step Poisson argument that passes back through Möbius sums and shrinks the number of rows each time, until the problem becomes trivial.

Scope. This is a guide to the argument of [1], specialized to ζ, not a new independent proof of its analytic estimates. Sections 1–5 give the deduction from one precisely stated mean-square estimate in full. Sections 6–8 explain the estimates behind it, retaining the parameters needed to see how the argument closes. The cubic theta theorem and the quadratic large sieve are substantial inputs.

1 The ordinary integer coefficients

Let μ be the ordinary Möbius function and put

$$\chi_{-3}(m) = \begin{cases} 0, & 3 \mid m, \\ 1, & m \equiv 1 \pmod 3, \\ -1, & m \equiv 2 \pmod 3. \end{cases}$$

The coefficients of the reciprocal product are completely classical:

$$c(m) = \sum_{de=m} \mu(d)\mu(e)\chi_{-3}(e), \qquad \frac{1}{\zeta(s)L(s,\chi_{-3})} = \sum_{m\geq 1} \frac{c(m)}{m^s} \quad (\Re s > 1). \tag{1}$$

This follows by multiplying the two absolutely convergent reciprocal Dirichlet series. For primes other than 3, the local coefficients are

	$c(p)$	$c(p^2)$	$c(p^j),\ j \geq 3$
$p \equiv 1 \pmod 3$	-2	1	0
$p \equiv 2 \pmod 3$	0	-1	0

and c is multiplicative. Thus c differs from μ: for example, $c(7) = -2$, $c(49) = 1$, and $c(25) = -1$.

Choose a complex-valued smooth function $W \in C_c^\infty((0,\infty);\mathbb{C})$ supported in a fixed interval $[a,b] \subset (0,\infty)$. Think of $W(m/D)$ as a smooth way of restricting m to size D. The sum to be bounded is

$$A_1(D) = \sum_{\substack{m\geq 1 \\ (m,6)=1}} c(m)W(m/D). \tag{2}$$

Removing the primes 2 and 3 gives the exact identity

$$\sum_{(m,6)=1} \frac{c(m)}{m^s} = \frac{1}{(1-3^{-s})(1-4^{-s})\zeta(s)L(s,\chi_{-3})}. \tag{3}$$

At 2, the reciprocal Euler factor is $1 - 2^{-2s} = 1 - 4^{-s}$; this explains the 4 rather than 2 in this formula.

2 Why the sums use Eisenstein integers

Put $\omega = e^{2\pi i/3}$, so $\omega^2 + \omega + 1 = 0$. An Eisenstein integer is a point $n = a + b\omega$ with $a, b \in \mathbb{Z}$, and

$$\mathrm{N}(n) = |n|^2 = a^2 - ab + b^2.$$

These points form a triangular lattice in the plane. The ring $\mathbb{Z}[\omega]$ has unique factorization. Its six units are $\pm 1, \pm\omega, \pm\omega^2$. To avoid counting six associates, we choose the *primary* generator:

$$n \equiv 1 \pmod 3, \quad \text{equivalently } a \equiv 1 \pmod 3,\ b \equiv 0 \pmod 3.$$

Every element coprime to 3 has exactly one primary associate. In sums below, n, f, b_1 and divisors of primary elements are primary and coprime to 6, unless stated otherwise. The row variables u, k, h run over all of $\mathbb{Z}[\omega]$. A star means that the element is squarefree.

Define $\mu_K(n) = 0$ if a prime square divides n, and $\mu_K(n) = (-1)^r$ if n is a product of r distinct primary primes; $\mu_K(1) = 1$. This is the same factorization rule as for the ordinary Möbius function, applied in $\mathbb{Z}[\omega]$.

A rational prime $p \equiv 1 \pmod 3$ splits into two conjugate prime factors of norm p. A prime $p \equiv 2 \pmod 3$ remains prime, of norm p^2. Consequently

$$\sum_{\substack{n \text{ primary} \\ \mathrm{N}(n)=m}} \mu_K(n) = c(m) \qquad ((m,6)=1). \tag{4}$$

For a split prime the local factor is $(1-p^{-s})^2$; for an inert prime it is $1-p^{-2s}$. These are exactly the factors in (1). Thus (2) can also be written

$$A_1(D) = \sum_n \mu_K(n) W(\mathrm{N}(n)/D).$$

The two-dimensional notation keeps track of the two primes above a split rational prime separately. The norm alone loses this information.

3 The average, written out

For a prime π coprime to 6, the number $\mathrm{N}(\pi) - 1$ is divisible by 6. Define the sextic residue symbol by

$$\left(\frac{u}{\pi}\right)_6 = \text{the sixth root of unity congruent to } u^{(\mathrm{N}(\pi)-1)/6} \pmod \pi$$

when $\pi \nmid u$, and define it to be 0 when $\pi \mid u$. Extend it multiplicatively in the denominator and write $\chi_n(u) = (u/n)_6$. It is multiplicative in the numerator as well. For $n = 1$ its value is 1, including at $u = 0$. Whenever we take a power of a character, its zero values remain zero, also for negative or zero exponents.

The individual sums, and their mean square, are

$$A_u(D) = \sum_n \mu_K(n)\chi_n(u) W(\mathrm{N}(n)/D), \tag{5}$$

$$M(D,H) = \sum_{0<\mathrm{N}(u)\le H} |A_u(D)|^2. \tag{6}$$

There are about H rows u, and each row contains about D possible columns n. More explicitly, with $Q(a,b) = a^2 - ab + b^2$,

$$M(D,H) = \sum_{\substack{x,y\in\mathbb{Z} \\ 0<Q(x,y)\le H}} \left| \sum_{\substack{a\equiv 1 \ (3),\ b\equiv 0 \ (3) \\ (a+b\omega,2)=1}} \mu_K(a+b\omega) \left(\frac{x+y\omega}{a+b\omega}\right)_6 W\left(\frac{Q(a,b)}{D}\right) \right|^2. \tag{7}$$

This is an ordinary sum over integer coordinates. Coprimality to 2 means that a, b are not both even.

Expanding the square gives the requested double sum over the columns:

$$\boxed{M(D,H)=\sum_{n_1,n_2}\mu_K(n_1)\mu_K(n_2)\overline{W(\mathrm{N}(n_1)/D)}W(\mathrm{N}(n_2)/D)S_H(n_1,n_2),} \tag{8}$$

where the finite character correlation is

$$S_H(n_1,n_2)=\sum_{0<\mathrm{N}(u)\le H}\overline{\left(\frac{u}{n_1}\right)_6}\left(\frac{u}{n_2}\right)_6. \tag{9}$$

Every coefficient in this average is now specified by factorization, finite-field exponentiation, and a smooth weight.

For completeness, one can group every row by ordinary integer norms:

$$c_u(m)=\sum_{\substack{n\ \text{primary}\\ \mathrm{N}(n)=m}}\mu_K(n)\chi_n(u),\qquad A_u(D)=\sum_{(m,6)=1}c_u(m)W(m/D).$$

Here $c_1(m)=c(m)$ from (1). In general $c_u(m)$ requires the residue symbols: μ and χ_{-3} alone do not specify the separate twists at conjugate primes.

The analytic estimate to prove. For every fixed $0<\vartheta\le 1/10$ and every $\varepsilon>0$, the required estimate is

$$\boxed{M(D,D^{1+\vartheta})\ll_{W,\vartheta,\varepsilon}D^{2+\vartheta+\varepsilon}.} \tag{10}$$

The notation $\ll$ means an inequality up to a constant independent of D. The factor D^{ε} allows an arbitrarily small power loss. The full argument controls the constant using finitely many derivatives of W. The estimate says that the *average* squared row sum is at most $D^{1+\varepsilon}$; it does not assert such a bound for every row.

4 The diagonal and the special rows

4.1 Why HD is the natural scale

For $n_1=n_2=n$, the correlation is simply

$$S_H(n,n)=\#\{u\in\mathbb{Z}[\omega]:0<\mathrm{N}(u)\le H,\ (u,n)=1\}.$$

For fixed n, elementary lattice counting and inclusion–exclusion give

$$S_H(n,n)=\frac{2\pi}{\sqrt{3}}H\prod_{\pi\mid n}\left(1-\frac{1}{\mathrm{N}(\pi)}\right)+O\big(\tau(n)(\sqrt{H}+1)\big),$$

where $\tau(n)$ is the number of primary divisors of n. The area of the disc is πH and the lattice has covolume $\sqrt{3}/2$. For the proof, the simpler uniform upper bound $S_H(n,n)\ll H$ suffices. The diagonal contribution to (8) is therefore $O_W(HD)$. Estimate (10) bounds the whole mean square on this scale, up to D^{ε}. It controls the aggregate off-diagonal contribution; it does not bound the sum of the absolute values of individual off-diagonal terms.

4.2 Sixth powers expose the untwisted sum

If $u = v^6$, then

$$\chi_n(v^6) = \mathbf{1}_{(v,n)=1}, \qquad A_{v^6}(D) = \sum_{(n,v)=1} \mu_K(n) W(\mathrm{N}(n)/D).$$

We use prime $v = \pi$ of comparable size so that the omitted terms are few. Put $Y = H^{1/6}$ and

$$\mathcal{P}_Y = \{\pi \text{ primary prime}, \ \pi \nmid 6 : Y/2 < \mathrm{N}(\pi) \le Y\}, \qquad J = |\mathcal{P}_Y|.$$

Then $\mathrm{N}(\pi^6) \le H$, and

$$|A_{\pi^6}(D) - A_1(D)| \ll_W D/Y. \tag{11}$$

Indeed, a contributing omitted n is πm with $\mathrm{N}(m) \le bD/\mathrm{N}(\pi)$; a disc of squared radius T contains $O(1+T)$ lattice points. In our range $D/Y \ge 1$. The prime number theorem in the progression 1 (mod 3) gives $J \asymp Y/\log Y$. These rows are distinct: equal sixth powers have associated bases, and primary associates are unique.

The elementary inequality $|z|^2 \le 2|w|^2 + 2|z-w|^2$, followed by positivity of (6), gives the rigorous extraction

$$J|A_1(D)|^2 \le 2 \sum_{\pi \in \mathcal{P}_Y} |A_{\pi^6}(D)|^2 + O_W(JD^2/Y^2) \le 2M(D,H) + O_W(JD^2/Y^2). \tag{12}$$

The dyadic interval $Y/2 < \mathrm{N}(\pi) \le Y$ is useful: it makes the error D/Y uniform for every selected row.

4.3 Cubes, squares, and rational rows

There are two other useful exact identities:

$$\chi_n(v^3) = \left(\frac{v}{n}\right)_2, \qquad \chi_n(v^2) = \left(\frac{v}{n}\right)_3.$$

Thus cube rows are quadratic twists and square rows are cubic twists. These classes overlap, as sixth powers illustrate. For a nonzero *rational integer* v and an odd denominator n,

$$\left(\frac{v}{n}\right)_2 = \left(\frac{v}{\mathrm{N}(n)}\right),$$

with the ordinary Jacobi symbol on the right. For split primes this is Euler's criterion over $\mathbb{F}_p$; for an inert prime of norm p^2, every nonzero element of $\mathbb{F}_p$ is a square in $\mathbb{F}_{p^2}$. Both sides vanish if $p \mid v$. Consequently

$$A_{v^3}(D) = \sum_{(m,6)=1} c(m) \left(\frac{v}{m}\right) W(m/D). \tag{13}$$

If $\chi_v(m) = (v/m)$ on odd m denotes the corresponding quadratic Dirichlet character there, the Dirichlet series of these coefficients is

$$\prod_{p \nmid 6} (1 - \chi_v(p)p^{-s})(1 - \chi_v(p)\chi_{-3}(p)p^{-s}).$$

This states the Euler factors explicitly without conventions at 2.

A rational numerator does not generally eliminate the need for Eisenstein residue symbols in a sextic or cubic twist. Also, the rational slice $u \in \mathbb{Z}$, $0 < \mathrm{N}(u) \le H$, has $\asymp H^{1/2}$ rows, since $\mathrm{N}(u) = u^2$. The whole lattice has $\asymp H$ rows. Sixth-power lattice rows number $\asymp H^{1/6}$; sixth powers with rational bases number $\asymp H^{1/12}$. These are different counts.

5 From the mean square to the zero-free region

Theorem 1. *If (10) holds for every smooth compactly supported W and every fixed $0 < \vartheta \le 1/10$, then $\zeta(s)$ has no zeros in $\Re s > 11/12$.*

Proof. Take $H = D^{1+\vartheta}$ in (12). Absorbing a logarithm into D^{ε} gives

$$\begin{aligned} |A_1(D)|^2 &\ll_{W,\vartheta,\varepsilon} DH^{5/6}D^{\varepsilon} + D^2H^{-1/3} \\ &= D^{11/6+5\vartheta/6+\varepsilon} + D^{5/3-\vartheta/3}. \end{aligned}$$

Taking square roots, and then choosing ϑ and ε sufficiently small, yields, for every $\eta > 0$,

$$A_1(D) \ll_{W,\eta} D^{11/12+\eta}. \tag{14}$$

The exponent $11/12$ is half of $2 - 1/6$: the mean square has size roughly D^2, and the useful rows number roughly $D^{1/6}$.

Here is the precise link with zeros. Write

$$F(s) = (1-3^{-s})(1-4^{-s})\zeta(s)L(s,\chi_{-3}), \qquad \widehat{W}(s) = \int_0^\infty W(y)y^s\,\frac{dy}{y}.$$

Suppose $\zeta(\rho) = 0$ and $\Re\rho > 11/12$. Choose a nonzero nonnegative smooth function ϕ supported in $(1,2)$ and set $W(y) = y^{-\rho}\phi(y)$. Then $\widehat{W}(\rho) > 0$. By (14),

$$B_W(s) = \int_0^\infty A_1(D)D^{-s}\,\frac{dD}{D}$$

is holomorphic in $\Re s > 11/12$. There is no difficulty at $D = 0$: the sum vanishes for $D < 1/2$. For $\Re s > 1$, absolute convergence and (3) give $F(s)B_W(s) = \widehat{W}(s)$. Meromorphic continuation extends this identity to $\Re s > 11/12$. Since $L(s,\chi_{-3})$ is entire and $\rho \ne 1$, $F(\rho) = 0$. Evaluation at ρ now gives $0 = \widehat{W}(\rho) > 0$, a contradiction. □

The remaining question is how to prove (10). The next three sections explain the argument of [1], specialized to ζ, and identify its hard inputs.

6 Poisson summation changes the coefficients

6.1 A smooth row cutoff

Replace the disc cutoff by a fixed nonnegative smooth radial weight $\Phi(\mathrm{N}(u)/H)$ with $\Phi(t) \ge 1$ for $0 \le t \le 1$. One can choose Φ rapidly decreasing and its two-dimensional Fourier transform $\widehat{\Phi}$ compactly supported. This gives an upper bound for M. It is this smoothed average to which Poisson summation is applied.

Here is an exact version of the transformation. Put

$$\lambda = 1 + 2\omega = i\sqrt{3}, \qquad e(z) = \exp(4\pi i\,\mathrm{Im}\,z/\sqrt{3}), \qquad \widehat{\Phi}(\mathrm{N}(w)) = \frac{2}{\sqrt{3}}\int_{\mathbb{C}} \Phi(|z|^2)e(-zw)\,dx\,dy.$$

Only squarefree n_1, n_2 occur because of the Möbius factors. Write $n_i = gz_i$ with $g = (n_1, n_2)$. Then z_1, z_2 are coprime and

$$\overline{\chi_{n_1}(u)}\chi_{n_2}(u) = \mathbf{1}_{(u,g)=1}\psi(u), \qquad \psi = \overline{\chi_{z_1}}\chi_{z_2}.$$

The character ψ is primitive modulo z_1z_2. For a primitive character modulo q, set

$$\gamma(\psi) = \mathrm{N}(q)^{-1/2} \sum_{x \bmod q} \psi(x)e(x/q).$$

Inclusion–exclusion for $(u, g) = 1$ and lattice Poisson summation give

$$\begin{aligned} &\sum_{u\in\mathbb{Z}[\omega]} \Phi(\mathrm{N}(u)/H)\mathbf{1}_{(u,g)=1}\psi(u) \\ &\quad = \frac{H\gamma(\psi)}{\sqrt{\mathrm{N}(z_1z_2)}} \sum_{d|g} \frac{\mu_K(d)\psi(d)}{\mathrm{N}(d)} \sum_{h\in\mathbb{Z}[\omega]} \overline{\psi(h)}\widehat{\Phi}\left(\frac{H\mathrm{N}(h)}{\mathrm{N}(d)\mathrm{N}(z_1z_2)}\right). \end{aligned} \tag{15}$$

The zero frequency $h = 0$ occurs only when $z_1 = z_2 = 1$, exactly the diagonal. All other frequencies satisfy

$$\mathrm{N}(h) \ll \frac{\mathrm{N}(d)\mathrm{N}(z_1z_2)}{H} \ll \frac{D^2}{H\mathrm{N}(g)},$$

because $d \mid g$ and $\mathrm{N}(z_i) \asymp D/\mathrm{N}(g)$. Thus a row range of size $H > D$ becomes a frequency range of size at most a constant times $D^2/H < D$. The divisor d in the exact formula must be retained.

6.2 The Möbius–Gauss identity

For squarefree n put

$$\gamma_j(n) = \mathrm{N}(n)^{-1/2} \sum_{x \bmod n} \chi_n(x)^j e(x/n), \qquad \alpha(n) = \frac{n}{|n|}, \qquad G(n) = \overline{\chi_n(4)}\gamma_3(n).$$

The arithmetic identity that makes the method work is

$$\mu_K(n)\gamma_{-1}(n) = \chi_n(-1)G(n)^{-1}\overline{\alpha(n)}\gamma_2(n). \tag{16}$$

The right side contains a *cubic* Gauss sum, since $\chi_n^2 = (\cdot/n)_3$. The unit-modulus factor G depends only on the residue class of n modulo 72 (in fact modulo 4 here).

To see its effect on (8), let $\mathfrak{G}$ be the finite group of invertible classes modulo 72 that are 1 modulo 3. Expand the function $t \mapsto G(t^{-1})$ in its finite group characters: $G(t^{-1}) = \sum_\xi c_\xi\xi(t)$. With $a_\xi(n) = \overline{\alpha(n)}\gamma_2(n)\xi(n)$, reciprocity and (16) give, for coprime squarefree z_1, z_2,

$$\mu_K(z_1)\mu_K(z_2)\gamma(\overline{\chi_{z_1}}\chi_{z_2}) = \sum_\xi c_\xi a_\xi(z_1)\overline{a_\xi(z_2)}. \tag{17}$$

There are only finitely many ξ, independently of D. Formula (17) preserves the form of a mean square while replacing its Möbius coefficients by cubic Gauss coefficients.

After resolving the coprimality conditions and dividing the variables into ranges $F \le \mathrm{N}(f) < 2F$, the averages to be bounded are

$$\mathcal{E}(\mathcal{H}, X, F) = \frac{1}{XF} \sideset{}{^*}\sum_{F\le\mathrm{N}(f)<2F} \sum_{0<\mathrm{N}(k)\le\mathcal{H}} \left| \sideset{}{^*}\sum_n a_\xi(n)\chi_n(k)\chi_n(f)^4 W(\mathrm{N}(n)/X)\right|^2. \tag{18}$$

Here ξ, W are suppressed from the notation. Both f, n are squarefree. The fourth power in $\chi_n(f)^4$ has two sources. Writing e for the exclusion divisor in Poisson summation gives a twist $\chi_n(he^5)$.

Removing the coprimality condition between the two columns introduces a common divisor v and the identity $a_\xi(vm) = a_\xi(v)a_\xi(m)\chi_m(v)^4$. With $f = ev$ and $k = eh$,

$$he^5v^4 = k(ev)^4 = kf^4.$$

Thus the v-part comes from twisted multiplicativity, and the e-part from the Poisson exclusion divisor. If $B \geq 1$ is a further divisor-size parameter, the initial ranges are

$$X = \frac{D}{BF}, \qquad \Sigma := XF = \frac{D}{B}, \qquad \mathcal{H} \ll \frac{D^2}{HB^2}. \tag{19}$$

The needed bound is $\mathcal{E} \ll D^\varepsilon \Sigma$. The normalizations and divisor sums then recover (10); this is the first Poisson reduction.

7 Why cubic theta and a quadratic large sieve appear

7.1 Adding the cube indices

The cubic Gauss coefficients in (18) occur, after a fixed normalization, in the Fourier expansion of Kubota's cubic theta function. Its indices include cube factors, powers of λ, and units. The explicit coefficient formulas are due to Patterson and are recorded in [2, Section 5.1 and Appendix].

Here is the normalization needed for our sum. Restrict a Fourier index ℓ to those for which $\lambda^3\ell$ is primary and coprime to 6. Among the nonzero coefficients, this selects exactly $\ell = \lambda^{-3}nb_1^3$, with n squarefree and b_1 arbitrary, both primary and coprime to 6. The coefficient of the conjugated theta function at such an index is

$$d_0(\lambda^{-3}nb_1^3) = 3^{5/2}|b_1|\,\widetilde{g}(n) = 3^{5/2}|b_1|\,\overline{\left(\frac{\lambda}{n}\right)_3}\gamma_2(n).$$

Eisenstein's supplementary law says that $(\lambda/n)_3$ depends only on n mod 9. Multiplying the twist at $x = \lambda^3\ell = nb_1^3$ by $(\lambda/x)_3 = (\lambda/n)_3$ cancels this extra factor. It is absorbed into the same finite congruence-class bookkeeping used above. This selection applies to the original series; unit factors and varying powers of λ reappear in the reflected cusp expansions and are included in their estimates.

For fixed k, f, put $\Psi(n) = \xi(n)\chi_n(k)\chi_n(f)^4$. The convenient completed sum is

$$T(X;k,f) = \frac{1}{\sqrt{X}}\sideset{}{^*}\sum_n \sum_{b_1} \overline{\alpha(n)}\gamma_2(n)\Psi(n)\overline{\alpha(b_1)}^3\Psi(b_1)^3\mathrm{N}(b_1)^{1/2}W\left(\frac{\mathrm{N}(n)\mathrm{N}(b_1)^3}{X}\right). \tag{20}$$

The $b_1 = 1$ part is precisely $X^{-1/2}$ times the column sum in (18). All sums here are finite because W has compact support.

7.2 The transformation and its purpose

The theta function obeys a transformation law relating its values at rational translates to its values near one of three cusps. Expanding Ψ in finite additive characters makes that law applicable to (20). Differentiating in the horizontal variable removes the constant Fourier terms; Mellin transformation then gives a reflected sum. The constant terms would otherwise produce residues, so their removal is an essential step.

The important local calculation is especially simple to state. At a prime where the original twist is χ_π^j, the reflected coefficient contains, up to a factor of absolute value 1,

$$B_{\pi,j}(x) = \begin{cases} \chi_\pi(x)^{-j-2}, & j \notin \{0,4\}, \\ \mathrm{N}(\pi)^{-1/2}\chi_\pi(x)^{-2}, & j = 0, \\ \mathrm{N}(\pi)^{-1/2}(-1 + \mathrm{N}(\pi)\mathbf{1}_{\pi\mid x}), & j = 4. \end{cases}$$

Characters are extended by zero at multiples of their prime, also when an exponent is 0. In particular,

$$B_{\pi,1}(x) = \chi_\pi(x)^3 = \left(\frac{x}{\pi}\right)_2.$$

Thus the squarefree part of the row twist becomes *quadratic*. If c is the denominator of the rational translate in lowest terms, put $Q = \mathrm{N}(c)$. The reflected weight has argument $\mathrm{N}(\ell)X/Q^2$, so the effective dual norm range is $\mathrm{N}(\ell) \lesssim Q^2/X$. It is shorter than the original range X when $Q < X$. Rapid decay becomes useful when X/Q^2 is large: the nonzero frequencies have $\mathrm{N}(\ell) \geq 1/81$, so their weight arguments then grow uniformly. The inequality $Q^2 < X$ alone is not a quantitative negligibility estimate.

The quadratic large sieve says, in the form needed here,

$$\sum_{\substack{\mathrm{N}(k)\leq K \\ (k,6)=1}}^{*} \left| \sum_{\mathrm{N}(n)\leq U}^{*} \beta(n) \left(\frac{n}{k}\right)_2 \right|^2 \ll_\varepsilon (KU)^\varepsilon (K+U) \sum_n^{*} |\beta(n)|^2. \tag{21}$$

Here the generators are primary, the column n may be even, and the coefficients $\beta(n)$ are arbitrary. Both the row and column squarefree restrictions are essential for this bound. In particular, if square columns $n = b_1^2$ were allowed, then $(b_1^2/k)_2 = 1$ whenever $(b_1, k) = 1$, so coefficients supported on squares would reinforce each other instead of cancelling. This is the substantial large-sieve input; it is a specialization of the number-field quadratic large sieve of [3].

Applying (21) to the reflected sum, accounting for the local factors at primes dividing f, and then treating the square factors of k, gives the completed estimate

$$\boxed{\sum_{0<\mathrm{N}(k)\leq\mathcal{H}} |T(X;k,f)|^2 \ll D^\varepsilon \left(\mathcal{H} + \frac{\mathcal{H}^2\mathrm{N}(f)}{X}\right).} \tag{22}$$

Parameters are bounded by fixed powers of D, and a fixed finite smoothness norm of W is understood. The estimate is derived from the theta transformation and the large sieve, as in [1].

For clarity, the passage to all rows uses $k = usv^2$, with u a unit and s squarefree. The identity $\chi_n(v^2)\chi_n(f)^4 = \chi_n(fv^2)^4$ absorbs the square into f. The corresponding squarefree-row estimate allows this more general second argument. Summing over v costs $\sum_v \mathrm{N}(v)^{-2} < \infty$.

8 Removing the added cubes and closing the argument

One cannot obtain a bound for the $b_1 = 1$ part of (20) simply by discarding the other terms: cancellation may occur inside T. Instead use the exact Möbius inversion

$$\frac{1}{\sqrt{X}} \sum_n^{*} a_\xi(n)\chi_n(k)\chi_n(f)^4 W(\mathrm{N}(n)/X) = \sum_h \frac{\mu_K(h)\overline{\alpha(h)}^3 \Psi(h)^3}{\mathrm{N}(h)} T(X/\mathrm{N}(h)^3; k, f). \tag{23}$$

In this formula h is primary and coprime to 6. Substituting (20) and collecting the total cube index uses $\sum_{h\mid b_1} \mu_K(h) = \mathbf{1}_{b_1=1}$, proving the identity.

8.1 Small cubes are controlled by the completed estimate

Choose

$$H_c^3 = \min\left(X, \frac{X^2}{\mathcal{H}^2}\right), \qquad L_{b_1} = X/\mathrm{N}(b_1)^3.$$

For $\mathrm{N}(h) \le H_c$, estimate (22) at scale $X/\mathrm{N}(h)^3$ gives terms bounded by

$$\mathcal{H} + \frac{\mathcal{H}^2 F \mathrm{N}(h)^3}{X} \ll \Sigma, \qquad \Sigma = XF,$$

provided $\mathcal{H} \le \Sigma$. Cauchy–Schwarz costs only divisor sums and logarithms, which fit into D^ε. Expanding the remaining completed sums in (23) leaves the same original average at smaller column scales. The resulting cube-removal estimate is

$$\mathcal{E}(\mathcal{H}, X, F) \ll D^\varepsilon \left(\Sigma + \sup_{\substack{\mathrm{N}(b_1) > H_c \\ L_{b_1} > 1}} \mathcal{E}(\mathcal{H}, L_{b_1}, F) \right). \tag{24}$$

Scales $L_{b_1} \le 1$ are bounded by counting. If $\mathcal{H}^2 \le X$, the supremum is empty and the desired estimate already follows.

8.2 Large cubes force a smaller row range

For the remaining terms, two more Poisson summations exchange cubic Gauss coefficients for Möbius coefficients and then back again. The paired identity (17) is used in both directions. The first summation gives frequencies of norm about $L^2/\mathcal{H}$, with additional divisor factors. Combining the frequency with the auxiliary variable as $y = hf^2$ gives a natural Möbius-row range of size $L^2F^2/\mathcal{H}$. A crucial step is to enlarge this range by positivity before the second Poisson summation. Suppressing the common divisor factor $\mathrm{N}(d)/\mathrm{N}(C)^2$ and fixed support constants, the two scales are

$$Y_{\text{natural}} = \frac{L^2F^2}{\mathcal{H}}, \qquad Y_{\text{enlarged}} = \frac{\Sigma LF}{\mathcal{H}}, \qquad \frac{Y_{\text{enlarged}}}{Y_{\text{natural}}} = \frac{\Sigma}{LF} \ge 1.$$

After separating the smooth weights, the row sums are nonnegative mean squares, so the larger smooth majorant gives a valid upper bound. The enlarged range makes the second Poisson frequency range smaller. For a remaining divisor r in a range $R \le \mathrm{N}(r) < 2R$, it gives

$$\mathcal{H}' = \frac{\mathcal{H}L}{\Sigma F R^2}.$$

Without enlargement the corresponding scale would be $\mathcal{H}/(F^2R^2)$, which need not decrease when $F = R = 1$.

After the second summation, regrouping the divisor terms *before* taking absolute values forces a divisibility condition $f' \mid k'$. At each prime the cancellation is

$$(1 + \mathbf{1}_{\pi \mid k'}) - \mathbf{1}_{\pi \mid k'} - \mathbf{1}_{\pi \nmid k'} = \mathbf{1}_{\pi \mid k'}.$$

This divisibility keeps the next average in the ranges needed for induction. If $F' \le \mathrm{N}(f') < 2F'$, then $F' \le \mathrm{N}(k') \le \mathcal{H}'$, so $\mathcal{H}' \ge F' \ge 1$. It also keeps the column scales at least 1: with $X' = L/(RF')$, the scales $X'/\mathrm{N}(j)$ produced by removing an exclusion divisor $j \mid r$ satisfy $X'/\mathrm{N}(j) \ge \Sigma F/\mathcal{H} \ge 1$.

Precisely, the transfer estimate bounds an average with column scale L by $D^{\varepsilon}\Sigma$ times 1 plus normalized averages $\mathcal{E}(\mathcal{H}', X', F')/\Sigma'$, where $\Sigma' = X'F'$ and

$$\mathcal{H}' \le \frac{\mathcal{H}L}{\Sigma F}, \qquad \frac{\mathcal{H}'}{\Sigma'} \le \frac{\mathcal{H}}{\Sigma}, \qquad \Sigma' \le L. \tag{25}$$

All three new parameters are at least 1; an empty collection contributes zero. Finite residue-class twists and smooth weights are allowed to change, with controlled derivatives. In (25) we record the exact scale inequalities of this transfer step, which are what the induction needs.

Suppose now that $\mathcal{H} \le \Sigma D^{-\kappa}$ for a fixed $\kappa > 0$. In the nonempty supremum of (24), $\mathrm{N}(b_1)^3 > X^2/\mathcal{H}^2$. Apply (25) with $L = L_{b_1}$. Then

$$\mathcal{H}' \le \frac{\mathcal{H}}{\mathrm{N}(b_1)^3 F^2} < \mathcal{H}\left(\frac{\mathcal{H}}{XF}\right)^2 \le \mathcal{H}D^{-2\kappa}. \tag{26}$$

The row range decreases by a fixed power of D, and the ratio condition $\mathcal{H}'/\Sigma' \le D^{-\kappa}$ survives. With all parameters bounded by D^{C_0}, finitely many induction steps reach $\mathcal{H} \le 1$. There are then at most six rows, and elementary counting gives $\mathcal{E} \ll X \le \Sigma$.

One formalizes this by induction on the ranges $\mathcal{H} \le D^{j\kappa}$. The required number of derivatives increases at each step, but the number of steps is bounded in terms of C_0, κ. Dividing the allowed ε loss among those steps proves

$$\mathcal{E}(\mathcal{H}, X, F) \ll D^{\varepsilon}\Sigma \qquad (\mathcal{H} \le \Sigma D^{-\kappa}). \tag{27}$$

Finally, the initial scales (19), with $H = D^{1+\vartheta}$, satisfy

$$\frac{\mathcal{H}}{\Sigma} \ll \frac{D^{-\vartheta}}{B}.$$

For sufficiently large D this is at most $D^{-\vartheta/2}$. Thus (27) applies with $\kappa = \vartheta/2$. Empty ranges and bounded column scales are handled by counting. The first Poisson reduction gives (10), and Section 5 then gives the zero-free half-plane.

The roles of the ingredients. The ordinary convolution $c = \mu * (\mu\chi_{-3})$ identifies the reciprocal zeta product. Sixth-power rows recover that convolution inside a larger nonnegative average. Poisson summation shortens the row range and exposes cubic Gauss sums. The theta transformation converts their twists into quadratic ones, to which a large sieve applies. Cube inversion and the contraction (26) transfer the resulting completed bound back to the original sums.

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